arXiv:2606.31650v5 [cs.LG] 3 Aug 2026

ECHO: PRUNE TO ACT, TRACE TO LEARN WITH SELECTIVE TURN MEMORY IN AGENTIC RL

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ABSTRACT

Long-horizon language agents must repeatedly interact with tools, accumulate evidence, and make decisions under bounded context windows. Context-management methods make such rollouts feasible by simplifying past interactions through deletion, folding, or memory editing. However, when useful history is collapsed into compressed states, the reconstructed context may no longer reveal which earlier observations support a successful final answer. This creates a mismatch between bounded-context acting and outcome-based reinforcement learning: the policy acts on reconstructed context, while the learner lacks source-level provenance for assigning credit to the evidence that mattered. We propose ECHO, a selective turn-memory framework for traceable context reconstruction in agentic RL. ECHO compresses each completed environment turn into a compact source-indexed memory record, reconstructs bounded policy contexts by selecting useful records, and reuses the selected source indices to route positive outcome credit to the final trajectory segment, reused evidence turns, memory findings, and memory-selection actions. On BrowseComp-Plus, ECHO reaches 43.4% held-out accuracy, outperforming GRPO at 28.9% and the rolling-summary baseline SUPO at 36.1%, while using fewer turns and lower trajectory volume than SUPO. The trained policy also improves zero-shot generalization across multi-objective QA, code generation, and deep information-seeking benchmarks on both dense and MoE backbones.

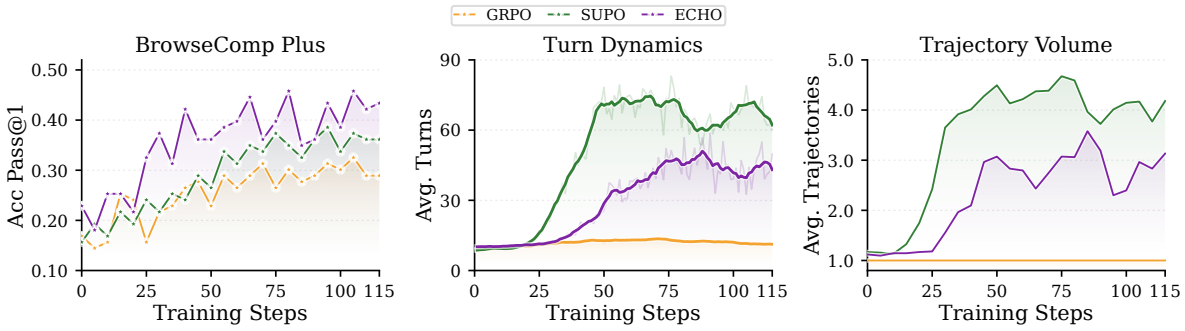


Figure 1: Held-out accuracy, tool-use turns per rollout, and trajectory volume over training on BrowseComp-Plus with the Qwen3-32B-Instruct backbone for ECHO (purple), GRPO (orange), and SUPO (green). ECHO traces the upper-left frontier: rising accuracy without the turn and volume growth seen for SUPO.

1 INTRODUCTION

Large language models (LLMs) are increasingly deployed as multi-turn agents that interleave reasoning, tool invocation, and environment feedback (Yao et al., 2023; Schick et al., 2023). Reinforcement learning (RL) from verifiable final outcomes has become a central recipe for improving such agents in search, coding, function calling, and deep-research settings (Jin et al., 2025; Qian et al., 2025; Li et al., 2025; Zheng et al., 2025). As interaction horizons grow, however, history management becomes a bottleneck for both acting and learning. The policy must retain useful observations within a bounded context window, while the learner must decide which earlier decisions should receive credit from a sparse final outcome.

Context-management methods make long rollouts feasible by simplifying past interactions before the next decision. From the perspective of reconstructed history, these methods differ in what form the completed history

becomes: some delete or omit distant turns, some edit an explicit memory state, and others fold long prefixes into compact summaries or internal states. The triggering mechanism may be rule-based or action-based, but this choice is orthogonal to the reconstructed form of history. For outcome-based RL, the central question is not only whether the shortened context is useful for acting, but also whether it remains traceable to the original environment turns for learning.

This provenance gap affects credit assignment in outcome-based agentic RL. With final-outcome rewards, standard multi-turn RL often applies a trajectory-level advantage to many generated action spans (Schulman et al., 2017; Shao et al., 2024). When the reconstructed context is not source-addressable, such dense assignment can reinforce redundant searches, incidental reasoning, or context-management behavior together with genuinely useful evidence-gathering steps. The issue is therefore not only whether context management enables longer bounded-context rollouts, but whether the reconstructed context exposes the source turns that the learner should credit. This suggests that a context manager should prune history for acting while preserving source-level provenance for learning.

In this paper, we propose ECHO, a selective turn-memory framework for traceable context reconstruction in agentic RL. Instead of collapsing distant history into a single summary state, ECHO compresses each completed tool-use turn into a compact memory record with a source-turn pointer. When the context budget becomes binding, the policy selects useful memory records and reconstructs the next bounded context from these selected records together with recent interactions. The agent therefore acts with compact context, while the selected memories remain explicitly linked to their original environment turns.

The same source-indexed reconstruction trace is then reused for learning. Standard trajectory-level updates attach the group-relative advantage, whether positive or negative, to all generated tokens in a rollout. In contrast, ECHO first keeps only the positive part of the trajectory advantage, and then applies it only to traceable credit tokens: the final trajectory segment, selected historical source turns, their memory findings, and the memory-selection actions that constructed the reconstructed context. Tokens outside this trace receive no positive outcome credit. In this way, ECHO changes credit assignment from dense trajectory-wide supervision to provenance-guided positive credit routing, aligning context pruning for acting with outcome-credit routing for learning through a single source-indexed trace. Together, these design choices make context reconstruction not only a way to fit long rollouts into a bounded context, but also a learning interface for source-aware credit assignment.

Our main contributions are as follows:

- We formulate context-managed multi-turn RL with a unified reconstruction interface that links history transformation to source traceability. This view shows why context management can make long rollouts feasible while still removing the provenance needed for outcome-credit assignment.
- We introduce ECHO, a turn-memory method that stores each turn as a source-indexed memory record and reconstructs bounded policy contexts through learned memory selection rather than collapsed-history folding.
- We propose provenance-guided token-level credit assignment, which reuses source indices to route positive outcome advantage to the final trajectory segment, historical evidence turns, memory findings, and memory-selection actions.
- Experiments on BrowseComp-Plus (Chen et al., 2025) and diverse zero-shot benchmarks show that ECHO improves accuracy while reducing turn proliferation and trajectory volume relative to the closest collapsed-history baseline, with consistent gains across dense and MoE backbones.

2 Related Work

2.1 RL for Long-Horizon Language Agents

Language agents are commonly formulated as multi-turn decision makers that interleave reasoning, tool invocation, and environment feedback (Yao et al., 2023). Recent RL methods improve such agents with outcome-based or verifiable rewards in search, function calling, coding, and deep-research environments (Schulman et al., 2017; Shao et al., 2024; Guo et al., 2025; Jin et al., 2025; Song et al., 2025; Li et al., 2025; Qian et al., 2025; Zheng et al., 2025; Du et al., 2025). These settings make context management necessary during rollout, because useful observations may span many tool-use turns and exceed the policy context budget. However, most agentic RL objectives treat the reconstructed policy context as given, and do not specify whether the information reused by the policy remains traceable to the original environment turns. ECHO addresses this missing interface between long-horizon rollout construction and learning by making reconstructed context source-indexed.

2.2 Context Management and Agent Memory

Prior work addresses long-context limitations through compression, summarization, retrieval, and memory modules (Li et al., 2023; 2024; Wang et al., 2024; Xu et al., 2024; Shen et al., 2025; Yang et al., 2025; Packer et al., 2023; Zhong et al., 2024; Chhikara et al., 2025; Xu et al., 2025). These methods improve bounded-context acting by shortening or externalizing past interactions, but they usually do not define how final outcome rewards should be linked back to the original turns that supplied useful evidence.

Recent agentic RL methods make context management trainable. Some replace the distant prefix with updated summaries or compact states (Wu et al., 2025; Lu et al., 2025; Yu et al., 2026; Zhou et al., 2025), while others expose memory or context editing as actions (Yan et al., 2025; Zhang et al., 2025; Ning et al., 2026; Sun et al., 2025). Under the history-transformation view in Table 1, these methods differ in whether history is deleted, edited, folded, or kept source-indexed. ECHO differs by storing each completed turn as a source-indexed memory record and reusing the selected records both for context reconstruction and outcome-credit routing. This taxonomy motivates our empirical focus on SUPO-style rolling summarization: it is the closest collapsed-history RL counterpart to ECHO under the same agent-loop reconstruction setting, isolating the difference between collapsed folding and source-indexed selection.

2.3 Credit Assignment in Agentic RL

Credit assignment is challenging when rewards are sparse and observed only after extended interactions (Williams, 1992; Sutton & Barto, 2018). Many outcome-supervised RL methods reuse a trajectory- or response-level advantage across generated action tokens or spans (Schulman et al., 2017; Shao et al., 2024; Yu et al., 2025). This dense assignment is simple, but noisy in long-horizon tool-use tasks where only a small subset of searches, observations, and intermediate decisions supports the final answer.

Recent methods refine credit along temporal, hindsight, milestone, hierarchical, or uncertainty-based axes (Zeng et al., 2025; Tan et al., 2026; Wang et al., 2026; Peng et al., 2026; Zhang, 2026; Zhao et al., 2026). These approaches improve how rewards are distributed over time or policy structure, but are mostly independent of how the agent's effective context was reconstructed. ECHO is complementary: rather than estimating a new temporal advantage, it uses the source indices exposed by context reconstruction to define a provenance-guided token mask for positive outcome credit.

3 Preliminaries

3.1 Multi-Turn GRPO for Agentic RL

Let x denote an initial task prompt. At interaction step t, the agent conditions on a policy context c_t and samples an action $a_t \sim \pi_\theta(\cdot \mid c_t)$. The action may contain reasoning tokens, a tool invocation, an internal context-management operation, or a final answer. If a tool is invoked, the environment returns an observation o_t; otherwise o_t may be null. The resulting multi-turn rollout is denoted by

$$\tau = (a_1, o_1, \ldots, a_T, o_T), \tag{1}$$

and receives a sparse final reward $R(\tau)$.

For each prompt, GRPO (Shao et al., 2024) samples a group of N complete trajectories $\{\tau^{(1)}, \ldots, \tau^{(N)}\}$. Let $r^{(n)} = R(\tau^{(n)})$, and let $\bar{r}$ and s_r be the mean and standard deviation of $\{r^{(i)}\}_{i=1}^{N}$. The group-relative advantage is

$$A^{(n)} = \frac{r^{(n)} - \bar{r}}{s_r + \epsilon}. \tag{2}$$

We use the following simplified policy-gradient form:

$$\mathcal{J}_{\mathrm{MT}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}} \left[\frac{1}{N} \sum_{n=1}^{N} \sum_{t=1}^{T_n} A^{(n)} \log \pi_\theta(a_t^{(n)} \mid c_t^{(n)}) \right]. \tag{3}$$

This expression suppresses implementation-specific surrogate terms, such as old-policy ratios, clipping or gating, token normalization, and optional KL penalties. Its purpose is to make explicit that a trajectory-level advantage is attached to generated actions conditioned on the policy contexts actually provided to the model. In token-level implementations, each action a_t is expanded into generated tokens, and the corresponding token log probabilities share the same trajectory-level advantage.

Here c_t denotes the bounded policy context used for the current model call. Standard multi-turn RL objectives typically treat this context as given. The next subsection makes the context-management mechanism explicit by separating the complete environment-side history from the reconstructed policy context.

Table 1: Context-managed rollout strategies under the $(\mathcal{M}, \Phi)$ interface. We organize methods by how they transform completed history. Superscripts on examples indicate how reconstruction is triggered: R denotes rule-triggered reconstruction, and A denotes action-triggered reconstruction. Source trace indicates whether the reconstructed context remains addressable at the level of original environment turns.

History Transformation	**Managed State**	**Reconstruction Form**	**Source Trace**	**Examples**
Append-only	H_j^{pre}	$H_j^{\mathrm{pre}} \oplus H_{j,t}^{\mathrm{loc}}$	Explicit	Vanilla prompting
Deletion / pruning	$C_j \subseteq H_j^{\mathrm{pre}}$	$\mathrm{Render}(C_j, H_{j,t}^{\mathrm{loc}}; B)$	Partial / lost	Sliding window^{R}, Agent-Omit^{A}
Edited memory state	B_j	$\mathrm{Render}(B_j, H_{j,t}^{\mathrm{loc}}; B)$	Action-traced	MemAct^{A}, Memory-R1^{A}
Collapsed folding	$z_j = C(z_{j-1}, \sigma_{j-1})$	$\mathrm{Render}(z_j) \oplus H_{j,t}^{\mathrm{loc}}$	Collapsed	SUPO^{R}, FoldGRPO^{A}
Selective turn memory	$\mathcal{E}_j = \{e_i\}_{i \le K_{j-1}}$	$\mathrm{Render}(\mathcal{E}_j[\hat{I}_j]) \oplus H_{j,t}^{\mathrm{loc}}$	**Source-indexed**	**ECHO**^{R}

3.2 Context-Managed Multi-Turn Rollouts

In long-horizon tool-use tasks, the complete environment-side history may exceed the context budget used during rollout and training. We write one completed interaction turn as $u_t = (a_t, o_t)$, so that the full rollout can be equivalently written as $\tau = (u_1, \ldots, u_T)$. Let B be the maximum policy context length. A context-managed rollout must construct a policy context c_t satisfying $|c_t| \le B$ at every interaction step.

For clarity, we first describe one trajectory and omit the trajectory superscript. We partition the rollout into bounded-context segments:

$$0 = K_0 < K_1 < \cdots < K_J = T. \tag{4}$$

The boundaries $\{K_j\}$ may be triggered by rule-based controllers, such as length or turn-count conditions, or by policy-generated context-management actions. The turns in segment j are indexed by $\mathcal{T}_j = \{K_{j-1}+1, \ldots, K_j\}$. For segment j, the completed prefix and the local within-segment history before turn t are

$$H_j^{\mathrm{pre}} = (u_1, \ldots, u_{K_{j-1}}), \quad H_{j,t}^{\mathrm{loc}} = (u_{K_{j-1}+1}, \ldots, u_{t-1}), \quad K_{j-1} < t \le K_j. \tag{5}$$

A context manager maps the completed prefix into a managed state, and a reconstruction function renders the bounded policy context:

$$z_j = \mathcal{M}(H_j^{\mathrm{pre}}), \quad c_{j,t} = x \oplus \Phi\big(z_j, H_{j,t}^{\mathrm{loc}}; B\big). \tag{6}$$

Here z_j may be retained history, a summary, a compressed state, an edited memory, or a set of memory units. The function Φ combines z_j with the local history under the budget B, and $\oplus$ denotes prompt concatenation, or more generally the effective conditioning mechanism.

This interface separates two properties of a context manager: whether it keeps the policy context short enough for acting, and whether the reconstructed information remains source-addressable for learning. Traceable methods may expose metadata such as source indices of retained or selected turns, which can later be used for credit assignment.

With context reconstruction, the reward and group-relative advantage remain unchanged. What changes is only the conditioning path from completed history to policy input:

$$H_j^{\mathrm{pre}} \xrightarrow{\mathcal{M}} z_j \xrightarrow{\Phi(\cdot, H_{j,t}^{\mathrm{loc}}; B)} c_{j,t}.$$

At the objective level, this simply replaces the original policy context $c_t^{(n)}$ in Eq. 3 with the reconstructed segment context $c_{j,t}^{(n)}$. For trajectory n, let $\mathcal{T}_j^{(n)}$ denote the turn indices of segment j. Then

$$\mathcal{J}_{\mathrm{CM}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}} \left[\frac{1}{N} \sum_{n=1}^{N} \sum_{j=1}^{J_n} \sum_{t \in \mathcal{T}_j^{(n)}} A^{(n)} \log \pi_\theta \left(a_t^{(n)} \mid c_{j,t}^{(n)} \right) \right]. \tag{7}$$

The segment index is only bookkeeping: the segments partition the same rollout, and the learning signal remains the trajectory-level advantage $A^{(n)}$. Thus, different context managers differ not by the reward estimator above, but by their choices of $\mathcal{M}$ and Φ, which determine the reconstructed context and its source trace. The key question is whether this trace remains addressable at the level of original environment turns, so that outcome credit can be routed to the evidence that the policy actually reused.

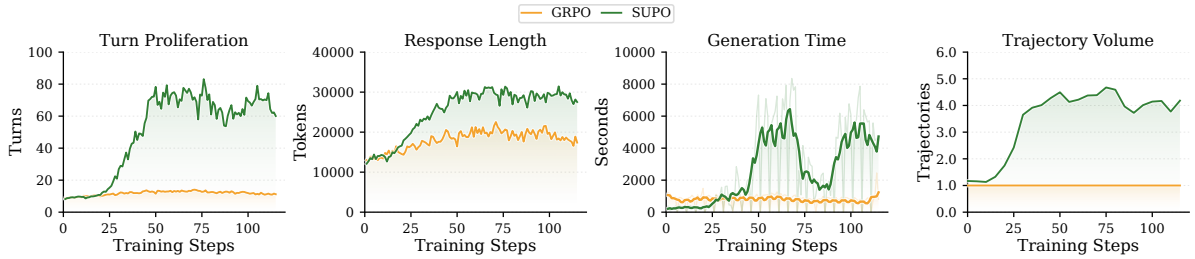


Figure 2: Training diagnostics on long-horizon search. Summarization-based context management enables longer rollouts, but leads to turn proliferation, longer responses, higher generation time, and increased trajectory volume.

4 METHOD

4.1 WHY TRACEABILITY MATTERS

The reconstruction interface in Eq. 6 separates two questions that are often conflated in long-horizon agentic RL: whether the policy can act under a bounded context budget, and whether the bounded context remains traceable to original environment turns. Table 1 organizes context managers by how they transform completed history. Append-only prompting preserves explicit trace but exceeds the context budget; deletion and pruning remove turns; edited-memory methods maintain compact memory states; and collapsed-folding methods replace long prefixes with summaries or compact states. These transformations make long rollouts feasible, but differ in whether reconstructed context remains source-addressable for learning.

This distinction matters because outcome-based RL assigns credit after the final answer is verified. If the policy conditions on reconstructed context whose source turns are no longer addressable, the learner cannot tell which earlier observations supported the successful decision. A dense trajectory-level update may then reinforce useful evidence-gathering turns together with redundant searches, incidental reasoning, or summary-generation tokens. Thus, context efficiency alone is not enough: the reconstruction process should expose a trace of the evidence that was selected and reused.

Collapsed summarization illustrates the limitation of history compression alone. By folding distant interactions into a compact summary state, it allows the agent to continue beyond the raw-context limit. However, the resulting context no longer exposes which original turns supplied the reused evidence. Figure 2 shows the practical consequence: summarization-based context management extends rollouts, but also produces more tool-use turns, longer responses, higher generation time, and larger trajectory volume. This supports the concern above: making history shorter for acting does not necessarily make learning easier. A context manager should therefore not only compress history, but also preserve explicit source traces that can support outcome-credit assignment.

4.2 SELECTIVE TURN MEMORY

ECHO instantiates source-indexed reconstruction with selective turn memory. The goal is to keep the active policy context compact without folding the distant prefix into a single untraceable state. To do so, ECHO separates local turn compression from global context reconstruction: each completed turn is compressed independently into a source-indexed memory record, and the next policy context is reconstructed by selecting useful records from the resulting memory set.

Source-indexed memory. After each completed tool-use turn $u_i = (a_i, o_i)$, ECHO asks the policy to write a compact local finding:

$$s_i \sim \pi_\theta(\cdot \mid c_i^{\text{sum}}), \qquad m_i = \text{Parse}(s_i). \tag{8}$$

Here c_i^{sum} contains the latest completed turn and a summary instruction that asks the model to summarize only that turn. This finding is generated by the same policy being trained, rather than by an external summarizer or a separate memory model. The memory record is

$$e_i = (i, \alpha_i, m_i), \qquad \mathcal{E}_j = \{e_i\}_{i \le K_{j-1}}. \tag{9}$$

The index i is the source-turn pointer, α_i is a compact rendering of the action or tool call, and m_i is the parsed last-turn finding. Unlike a rolling summary, $\mathcal{E}_j$ is a persistent, non-collapsing archive. Selection changes only the active view rendered into the next policy context; it does not delete unselected records. Thus, every memory remains individually addressable at later reconstruction boundaries even after its raw observation is no longer placed in the active context.

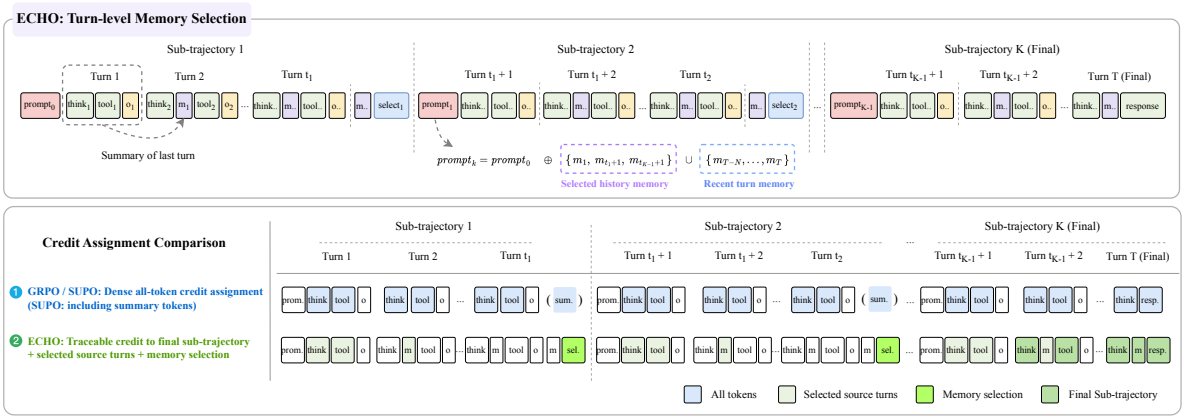


Figure 3: **Overview of ECHO.** ECHO stores completed turns as source-indexed memories, selects useful memories for bounded context reconstruction, and reuses the same source trace for credit assignment.

Memory selection. At a compression boundary before segment j, ECHO lets the policy select historical memories that are useful for continuing the task. Let H_j^{bd} denote the bounded local state available at the boundary. The selection context is

$$c_j^{\mathrm{sel}} = x \oplus \Phi_{\mathrm{local}}(H_j^{\mathrm{bd}}; B) \oplus \mathrm{RenderList}(\mathcal{E}_j) \oplus x_{\mathrm{sel}}, \tag{10}$$

and the policy generates a selection action

$$a_j^{\mathrm{sel}} \sim \pi_\theta(\cdot \mid c_j^{\mathrm{sel}}). \tag{11}$$

The instruction x_{sel} asks the model to output source indices of reusable evidence, constraints, failed attempts, or useful next-step plans. A repair operator converts the generated text into a valid bounded set:

$$\hat{I}_j = \rho_{B,S,K}\left(\mathrm{Parse}(a_j^{\mathrm{sel}}), \mathcal{E}_j\right). \tag{12}$$

The operator $\rho_{B,S,K}$ removes malformed or out-of-range indices, deduplicates selections, keeps at most S model-selected historical turns, and merges them with the latest K turns retained automatically. The rendered view is then validated against the context budget; a rollout is marked overlong if the bounded view still cannot be constructed. Thus, both memory writing and memory selection are policy-generated and trained within the same RL loop.

Context reconstruction. The selected indices determine which memories are used for reconstruction:

$$\mathcal{E}_j[\hat{I}_j] = \{e_i \in \mathcal{E}_j \mid i \in \hat{I}_j\}. \tag{13}$$

For any turn t in segment j, ECHO reconstructs the policy context as

$$c_{j,t} = x \oplus \mathrm{Render}(\mathcal{E}_j[\hat{I}_j]) \oplus \Phi_{\mathrm{local}}(H_{j,t}^{\mathrm{loc}}; B), \tag{14}$$

with $|c_{j,t}| \le B$. The agent acts on a compact context composed of selected memories and recent interactions. At the same time, the selected set $\hat{I}_j$ preserves a source-indexed reconstruction trace, which is reused for credit routing.

4.3 Traceable Credit Routing

ECHO uses a final-trace approximation for outcome-based learning: it routes credit through the source turns selected into the final reconstructed context, rather than recursively tracking all intermediate reconstruction dependencies. This final reconstruction trace defines which generated tokens are eligible for positive outcome credit.

For trajectory n, let $I_{\mathrm{src}}^{(n)}$ be the source turns selected into the final reconstructed context under this final-trace approximation. Let j index saved trajectory segments and q index a generated response-token position within segment j. For each token, we record four indicators: $g_{j,q}^{(n)}$ marks trainable policy tokens in the final saved segment, $d_{j,q}^{(n)}$ gives the source turn of a normal assistant/action token, $f_{j,q}^{(n)}$ gives the source turn of a generated memory-finding token, and $b_{j,q}^{(n)}$ marks memory-selection tokens. ECHO defines the hard credit mask

$$\mu_{j,q}^{(n)} = \mathbf{1}\left[g_{j,q}^{(n)} = 1 \vee d_{j,q}^{(n)} \in I_{\mathrm{src}}^{(n)} \vee f_{j,q}^{(n)} \in I_{\mathrm{src}}^{(n)} \vee b_{j,q}^{(n)} = 1\right]. \tag{15}$$

The mask marks tokens that are eligible for positive credit: all trainable tokens in the final trajectory segment, action tokens from selected source turns, their memory-finding tokens, and generated memory-selection spans.

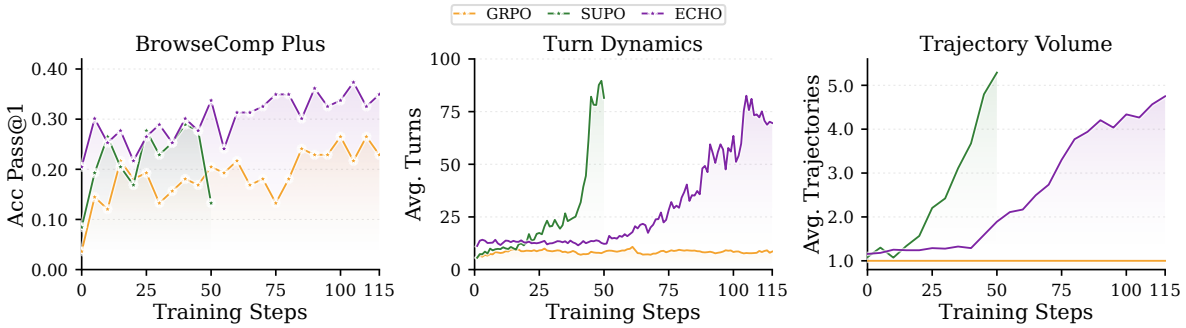


Figure 4: Training dynamics on BrowseComp-Plus with the Qwen3-30B-A3B-Instruct MoE backbone. The panels report held-out accuracy, average tool-use turns, and trajectory volume; ECHO improves accuracy without SUPO's turn and volume growth.

The final segment begins after the last context reconstruction and includes its assistant reasoning and tool-call tokens together with the terminal response.

For binary verifier rewards, the selected trace is reliable only when the rollout is better than the group baseline. If the rollout fails, selected turns may be irrelevant, misleading, or useful but misused. We therefore route only the positive part of the group-relative advantage:

$$A_+^{(n)} = \max(A^{(n)}, 0), \qquad \widetilde{A}_{j,q}^{(n)} = \mu_{j,q}^{(n)} A_+^{(n)}. \tag{16}$$

Because credit is multiplied by $A_+^{(n)}$, only positive-advantage rollouts assign nonzero credit to the masked tokens. Memory-selection tokens are included in the eligible set because they choose the memories that later successful decisions rely on. The per-trajectory traceable surrogate is

$$\mathcal{J}_{\mathrm{ECHO}}^{(n)}(\theta) = \sum_j \sum_q \widetilde{A}_{j,q}^{(n)} \log \pi_\theta \left(y_{j,q}^{(n)} \mid c_j^{(n)}, y_{j,<q}^{(n)} \right). \tag{17}$$

This surrogate has the same simplified policy-gradient interpretation as Eq. 3, but replaces dense trajectory-wide weighting with the trace-masked positive weight $\widetilde{A}_{j,q}^{(n)}$. Thus, ECHO turns source-indexed reconstruction into a token-level credit route.

5 Experiment

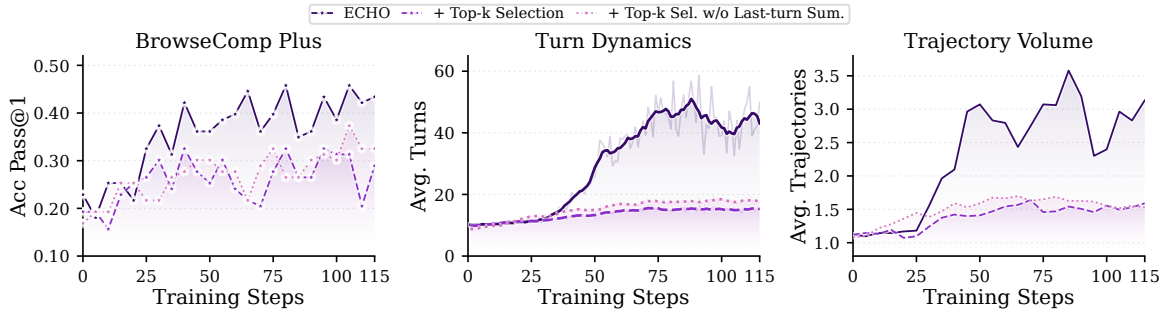
5.1 Experimental Setup

We use SUPO-style rolling summarization as the main context-managed RL baseline because it is the closest collapsed-history counterpart to ECHO. Both methods perform context reconstruction inside the agent loop and train the policy to generate the text used for context management, rather than relying on an external retriever or a separate memory model. The key difference is the form of reconstructed history: SUPO folds distant interactions into a collapsed summary state, whereas ECHO reconstructs context from selected source-indexed turn memories and reuses the selected source trace for credit routing.

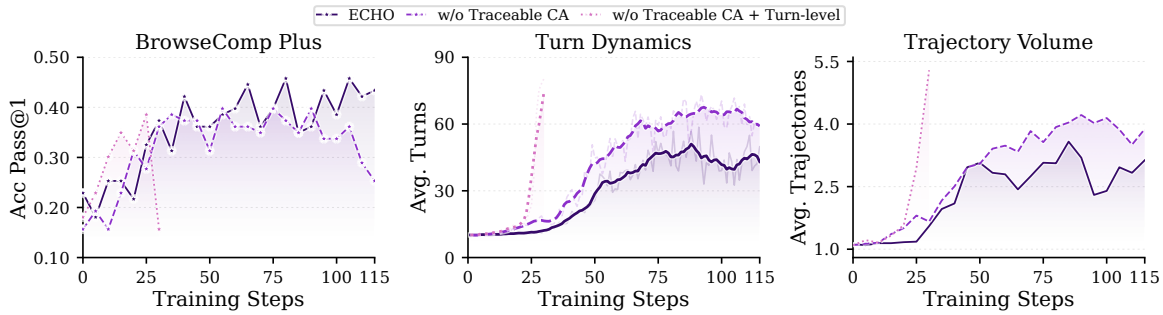
We evaluate on BrowseComp-Plus, a long-horizon tool-use QA benchmark, and compare GRPO (Shao et al., 2024), SUPO-style rolling summarization (Lu et al., 2025), and ECHO. All methods use the same search/open-page tool environment, verifier, 8 rollouts per prompt, and 32k-token working context budget. For context-managed methods, we allow at most 5 reconstruction rounds, which gives an effective interaction budget of up to 192k tokens while keeping each policy call within the 32k working context. ECHO automatically retains the latest 3 turns and selects up to 8 additional historical source turns at each reconstruction boundary. We report held-out pass@1 and rollout-efficiency metrics in the main text; dataset statistics, retrieval settings, and zero-shot benchmark details are provided in the appendix.

5.2 Main Results

Figure 1 shows that ECHO reaches 43.4% held-out accuracy on the dense backbone, outperforming GRPO (28.9%) and SUPO (36.1%). GRPO remains short but plateaus early, indicating that raw bounded-context training limits the amount of evidence the agent can gather. SUPO improves accuracy by extending rollouts, but this comes with substantial turn proliferation: its final turn count reaches 62.5, with an 85.5% trajectory split rate and 4.18 trajectories per rollout. ECHO surpasses SUPO while ending with fewer turns (45.3), a lower split rate



(a) Memory component ablation.



(b) Credit assignment ablation.

Figure 5: Ablations on memory components and credit assignment. Learned source selection is strongest, compact findings are sufficient, and removing traceable credit routing reduces accuracy and stability.

(57.8%), and fewer trajectories per rollout (3.13). These results suggest that ECHO's source-indexed selection and traceable credit routing help the agent reuse useful evidence more selectively, yielding more effective search paths under the same bounded-context setting.

Figure 4 shows the same ranking with the sparse Qwen3-30B-A3B-Instruct MoE backbone. GRPO reaches only 22.9% accuracy. SUPO initially benefits from rolling summaries, but its turn count and trajectory volume grow rapidly and accuracy drops to 13.3% by step 50. In contrast, ECHO remains stable and finishes around 35.0%. This suggests that source-indexed selection and traceable credit routing are not specific to the dense backbone.

5.3 Ablation Study

Memory ablations. Figure 5a studies how the memory component should be constructed and retrieved. Replacing learned source selection with static semantic top-k retrieval keeps the reconstructed context compact, but yields lower accuracy than full ECHO. This suggests that the benefit of turn memory does not come only from storing compressed historical records; the policy must also learn which records are useful for the current decision. Replacing compact turn findings with full observations also brings no clear gain, indicating that local findings are sufficient for preserving reusable evidence while avoiding unnecessary context growth. Together, these results support the design choice of storing compact source-indexed findings and performing policy-driven selection at reconstruction time.

Credit-routing ablations. Figure 5b evaluates whether the selected source trace should also guide credit assignment. Removing traceable credit assignment lowers accuracy and leads to longer interaction paths with more turns and larger trajectory volume. This suggests that source-indexed reconstruction alone is not sufficient: if the final outcome advantage is densely assigned to all generated tokens, the learner may still reinforce redundant searches, incidental reasoning, and other tokens that merely co-occur with a successful rollout. Adding turn-level importance sampling does not solve this issue; in our training curves, it becomes unstable and collapses as turn count and trajectory volume grow. In contrast, ECHO routes positive credit only through the final trajectory segment, selected evidence turns, their memory findings, and the selection actions that exposed those memories. This acts as a denoising mechanism for outcome credit and helps compress successful search paths rather than merely extending them.

Table 2: Zero-shot generalization results across various benchmarks. **Bold** denotes the best performance and underlined denotes the second-best. CA denotes credit assignment, Traceable CA denotes source-indexed credit routing, and Turn-level IS denotes importance-sampling ratios aggregated at the turn level.

Method	Multi-Objective QA					Code Generation		Deep Information Seeking			Avg.
	2-obj.	4-obj.	8-obj.	16-obj.	Avg.	CodeGym	LoCoBench-Agent	GAIA	HLE	Frames	
Backbone: Qwen3-32B-Instruct											
GRPO	38.6	39.8	35.8	29.0	35.8	32.8	67.7	25.2	8.8	24.8	33.6
SUPO	40.9	36.4	36.4	34.7	37.1	35.4	68.1	25.2	9.2	26.8	34.8
ECHO w/ Top-K retrieval	**47.7**	42.0	39.2	27.8	39.2	40.7	69.3	**29.1**	10.6	37.3	38.2
ECHO w/ Top-K retrieval & w/o turn summary	40.9	44.3	35.8	35.5	39.1	40.3	69.5	23.3	10.0	31.3	36.8
ECHO w/o traceable CA	45.5	42.0	39.2	22.7	37.4	38.1	68.2	25.2	8.8	30.8	35.6
ECHO w/o Traceable CA & Turn-level IS	45.5	**47.7**	35.2	27.0	38.8	34.6	70.1	23.3	9.4	32.2	36.1
ECHO	**47.7**	45.5	**41.5**	**36.1**	**42.7**	**41.4**	**70.4**	**29.1**	**11.4**	**39.1**	**40.2**
Backbone: Qwen3-30B-A3B-Instruct											
GRPO	27.3	26.1	27.3	16.2	24.2	20.3	65.7	23.3	7.8	19.1	25.9
SUPO	25.0	30.7	27.3	18.2	25.3	27.3	65.1	23.3	8.0	17.0	26.9
ECHO	**34.1**	**36.4**	**30.1**	**18.8**	**29.9**	**29.7**	**66.8**	**24.3**	**9.2**	**25.0**	**30.5**

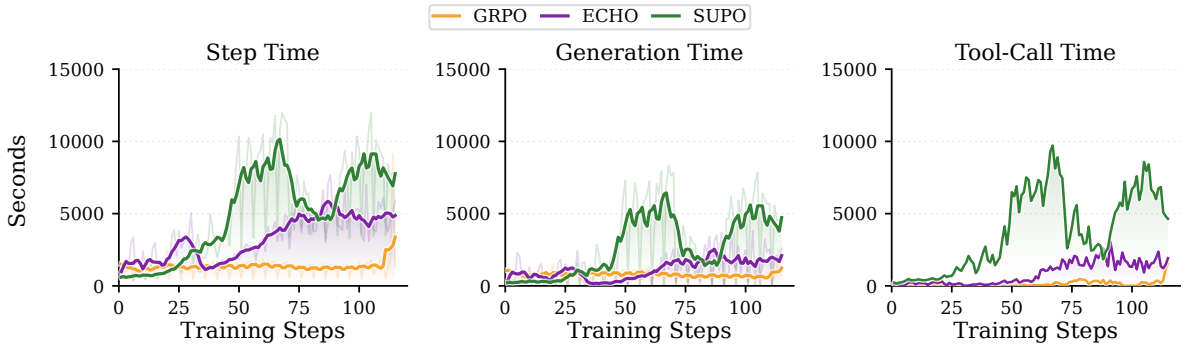


Figure 6: Training-time efficiency on BrowseComp-Plus. ECHO keeps step, generation, and tool-call time lower than SUPO.

5.4 Training Efficiency

Figure 6 compares wall-clock cost during dense-backbone training. ECHO introduces early overhead because each completed tool turn writes a `sum_last_turn` finding and compression boundaries add selection actions. As training proceeds, however, SUPO's rolling summaries lead to rapidly growing interaction rounds, making tool calls the dominant cost. By tying reconstruction to a bounded set of selected source turns, ECHO limits this mid-to-late-stage turn proliferation and achieves higher accuracy with fewer splits and lower tool-call-dominated execution time.

5.5 Zero-shot Generalization Comparison

The learned context-management behavior also transfers beyond BrowseComp-Plus when the trained policies are evaluated in the same agent-loop framework without benchmark-specific further training (Table 2). These results therefore measure zero-shot agentic transfer, rather than bare-model generation performance. On Qwen3-32B-Instruct, ECHO achieves the best average score of 40.2%, outperforming GRPO (33.6%) and SUPO (34.8%), with especially large gains on evidence-heavy 16-objective QA and Frames. Under the MoE backbone, ECHO reaches 30.5% average score versus 26.9% for SUPO. The zero-shot ablation rows follow the same trend as the BrowseComp-Plus training curves: replacing learned selection with semantic top-k retrieval lowers the dense-backbone average to 38.2%, and removing traceable credit further reduces it to 35.6%. These results suggest that the benefits of source-indexed reconstruction and traceable credit routing transfer across tasks and backbone architectures.

6 Conclusion

We presented ECHO, a selective turn-memory framework that stores each completed turn as a source-indexed memory, reconstructs compact contexts through learned selection, and reuses the same trace to route positive outcome credit to the final trajectory segment, reused evidence, memory findings, and selection actions. Under a

controlled agent-loop context-management setting, ECHO improves over the closest collapsed-history baseline, SUPO, while using fewer turns and lower trajectory volume. Ablations and zero-shot evaluations further show that learned source selection and traceable credit routing both contribute to robust transfer. These results suggest that long-horizon agentic RL benefits not only from shortening history, but from preserving a source-indexed trace of the history that the policy actually reuses. Looking ahead, we plan to extend ECHO from final-trace approximation to all-trace credit routing and train it on real software-engineering tasks with executable feedback, such as SWE-Bench. We will also compare its rule-triggered context management with action-based memory operations, including editing, omission, and folding, to further evaluate its generality.

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Algorithm 1 ECHO rollout agent loop

Require: Task prompt x, policy π_θ, tools $\mathcal{T}$, context budget B, recent-turn budget K, selection cap S

$C \leftarrow x$
$\mathcal{M} \leftarrow \emptyset$ ▷ persistent source-indexed archive
$\mathcal{S} \leftarrow \emptyset$; $G \leftarrow \emptyset$; $p \leftarrow \emptyset$; $I_{\text{final}} \leftarrow \emptyset$; $i \leftarrow 0$
while rollout is not terminated **do**
  Sample assistant response $a_t \sim \pi_\theta(\cdot \mid C)$
  Append trainable tokens of a_t to current segment G
  if $p \neq \emptyset$ **then**
    $(i_p, \alpha_{i_p}) \leftarrow p$
    Extract finding m_{i_p} from <sum_last_turn> in a_t
    $\mathcal{M} \leftarrow \mathcal{M} \cup \{(i_p, \alpha_{i_p}, m_{i_p})\}$
    Tag finding tokens in G with source id i_p
    $p \leftarrow \emptyset$
  end if
  Parse tool calls from a_t
  if no tool call remains **then**
    Mark all trainable tokens in G as the final segment
    Append $(G, \text{final} = 1)$ to $\mathcal{S}$
    break
  end if
  Execute tool calls and obtain observations o_t
  if committing o_t would exceed B **then**
    Roll back current a_t from G and discard its pending record
    Append $(G, \text{final} = 0)$ to $\mathcal{S}$
    Generate $a^{\text{sel}} \sim \pi_\theta(\cdot \mid c^{\text{sel}}(\mathcal{M}))$
    Append trainable tokens of a^{sel} to the saved segment and tag its <selection> span
    $I \leftarrow \rho_{B,S,K}(\text{Parse}(a^{\text{sel}}), \mathcal{M})$
    $I_{\text{final}} \leftarrow I$
    $C \leftarrow x \oplus \text{Render}(\mathcal{M}[I]) \oplus x_{\text{cont}}$
    Validate $|C| \leq B$; otherwise mark the rollout overlong
    $G \leftarrow \emptyset$
  else
    Tag action tokens of a_t in G with source id i
    $C \leftarrow C \oplus a_t \oplus o_t \oplus x_{\text{turn}}$
    $p \leftarrow (i, \text{RenderAction}(a_t))$; $i \leftarrow i + 1$
  end if
end while
Build credit masks from final-segment flags, I_{final}, source ids, and selection spans
return $\mathcal{S}$ with credit-token masks; keep $\mathcal{M}$ intact

A Experimental Details

Table 3 lists the dataset, rollout, retrieval, and context-management settings used in our experiments. The main text reports the settings most relevant to the controlled comparison, including the shared 32k-token working context, the maximum number of reconstruction rounds, and the resulting effective interaction budget. The remaining details specify the retrieval backend, tool limits, asynchronous rollout pipeline, and method-specific memory-selection budgets.

A.1 Evaluation Benchmarks

In this section, we provide detailed descriptions of the evaluation protocols and dataset characteristics for the zero-shot generalization experiments.

Multi-Objective QA. To evaluate the agent's long-range reasoning and context management capabilities under varying cognitive loads, we construct a multi-objective QA benchmark (Zhang et al., 2025). This benchmark challenges the agent to resolve multiple independent sub-questions within a single extended rollout to synthesize a final answer. We synthesize these multi-objective queries using seed instances from HotpotQA (Yang et al., 2018), 2WikiMultihopQA (Ho et al., 2020), Bamboogle (Press et al., 2023), and Musique (Trivedi et al., 2022). We scale the task difficulty by aggregating questions to form test sets requiring 2, 4, 8, and up to 16 distinct objectives. Performance is measured using an LLM-as-a-Judge (Zheng et al., 2023) protocol. The evaluator

Setting	Value
BrowseComp-Plus dataset	
Training examples	747
Held-out examples	83
Verifier	Deepseek V4-Flash LLM judge
Tool environment and retrieval	
Tools	`search`, `open_page`, `finish`
Search backend	Qwen3-Embedding-8B retrieval
Search return size	Top-5 documents
Search snippet length	2048 chars
`open_page` limit	16,000 chars
Parallel tool calls	At most 5 per assistant turn
Training and context management	
Rollout group size	8
Working context budget	32k tokens
Reconstruction rounds	At most 5
Effective interaction budget	Up to 192k tokens
Rollout pipeline	Fully asynchronous
Hardware	4 nodes, 8 GPUs per node
Partial-rollout staleness	0.5
Latest turns retained K	3
Selected historical turns S	At most 8

Table 3: Implementation settings for BrowseComp-Plus training and evaluation. The effective interaction budget counts the maximum cumulative history covered across reconstruction rounds; each individual policy call is still bounded by the 32k-token working context.

assesses the semantic consistency between the agent's final answer and the ground truth, computing the average success rate across all sub-objectives within a given multi-objective prompt.

Code Generation. We evaluate interactive programming capabilities using CodeGym (Du et al., 2025) and LoCoBench-Agent (Qiu et al., 2025). CodeGym frames tasks as synthetic interactive environments where agents invoke problem-specific APIs, together with `observe()` and `done()`, rather than writing raw code. Since many CodeGym tasks are either too easy or too difficult for meaningful comparison, we construct a medium-difficulty subset using the original `Qwen3-32B-Instruct` policy before agentic RL training. Specifically, we first run the base policy four times per instance, retain instances solved exactly once, and then randomly sample 128 examples from this candidate pool. This filtering is performed once before evaluating any trained method, and the same subset is used for all methods. For authentic software-engineering scenarios, we further evaluate on 128 Python instances from LoCoBench-Agent. These tasks operate on real codebases of up to 1 million tokens and require agents to manage cross-file dependencies, investigate bugs, and perform architectural refactoring with tools for file operations and semantic search.

Deep Information Seeking. Deep information seeking benchmarks measure the agent's proficiency in orchestrating complex reasoning, multi-step tool use, and long-horizon fact aggregation. We evaluate on GAIA (Mialon et al., 2024), which reflects real-world assistant tasks demanding rigorous execution and robust fact-checking. To test the limits of domain expertise, we use Humanity's Last Exam (HLE) (Phan et al., 2025), a frontier-level benchmark featuring expert-vetted questions that require deep reasoning rather than surface-level internet retrieval. Finally, we incorporate Frames (Krishna et al., 2025) to assess the agent's capacity to resolve multi-hop queries, integrate conflicting evidence, and maintain critical factual anchors across extensive retrieval trajectories without experiencing history collapse.

B Prompt Templates

We provide the prompt templates used by ECHO. Braces denote runtime-filled slots. The base BrowseComp-Plus prompt follows the same Hermes-style tool-call protocol as the SUPO baseline; ECHO adds the turn-summary and selection prompts below.

B.1 Base BrowseComp-Plus Prompt

Tool-Agent Prompt

You are an expert research agent tasked with answering the given question through iterative tool use.

Question: {question}

Available tools:

- search(query): retrieve candidate documents.
- open_page(docid): inspect a retrieved document.
- finish(answer): submit the final answer.

Follow this protocol:

1. Decompose the question into searchable facts.
2. Search for evidence and open useful pages.
3. Track constraints, aliases, dates, and sources.
4. When confident, call finish with a concise final answer.

Tool calls must use the Hermes-style format:

```
<tool_call>
{"name": "...", "arguments": {...}}
</tool_call>
```

B.2 ECHO-Specific Prompts and Hints

ECHO Extra System Prompt

After receiving a tool/function response, your next assistant message must include exactly one <sum_last_turn>...</sum_last_turn> block before any new tool/function call.
The block should be one concise factual sentence summarizing only the latest tool/function result.
After the block, continue in the same assistant message with the next tool/function call, or call finish if the final answer is ready. Do not stop after the summary.

Sum Last Turn Hint

Briefly record the latest tool result in <sum_last_turn>...</sum_last_turn>, then continue with the next action.

Selection Prompt

System:
Your operational context is full. Select prior interaction turns that are necessary to continue solving the task.

User:
Valid turns:
{turn_list}

Rules:

- Output exactly one <selection>...</selection> block.
- Do not call any function/tool in this turn.
- Do not answer the original task.
- Do not include <think>, tool calls, markdown fences, or text outside the selection tags.
- Select only turns that contain reusable evidence, constraints, failed attempts, or the next planned action.
- Prefer older turns that are not already covered by automatically retained recent turns.
- If no older turn is necessary, return an empty <selection></selection> block.

Output format:

```
<selection>[zero or more lines, each formatted as turn_N: reason]
</selection>
```

Selection Hint

Valid selection indices: `turn_0` through `turn_{n_hist-1}`.
Select at most `{selection_max_turns}` historical turns; output an empty selection if the automatically retained recent turns are enough.
The latest turns `turn_{recent_start}` through `turn_{n_hist-1}` will be retained automatically; select any additional older turns that are still needed.

In our BrowseComp-Plus experiments, `selection_max_turns = 8` and `echo_recent_turns = 3`.